

Fraïssé Theory of Random Graphs

Key facts from Fraïssé theory

To simplify things, we state everything in the context of a **finite language that contains only constants and relations**. In this case *finitely generated* is equivalent to *finite*.

- $\mathcal{M}$ is **homogeneous** if any isomorphism between finite substructures of $\mathcal{M}$ can be extended to an automorphism of $\mathcal{M}$.
- $\text{age}(\mathcal{M})$ is the class of all finite $\mathcal{L}$ -structures isomorphic to a substructure of $\mathcal{M}$.
- Any two countable homogeneous structures with the same age are isomorphic.
- A class of finite structures $\overline{K}$ is an **amalgamation class** if it satisfies (HP), (JEP), and (AP). In words, every substructure of a structure in $\overline{K}$ is in $\overline{K}$, any two structures in $\overline{K}$ embed into another structure from $\overline{K}$, and any two structures with a common substructure (up to isomorphism) amalgamate to another structure that preserves the common substructure.
- **Fraïssé's Theorem:** Suppose $\overline{K}$ is a class of finite $\mathcal{L}$ -structures such that there are only countably many isomorphism types in $\overline{K}$. Then $\overline{K}$ is an amalgamation class iff $\overline{K}$ is the age of a countable homogeneous $\mathcal{L}$ -structure.

Simple graphs

We consider the theory of *undirected graphs without self-loops*, also called **simple graphs**. Let $\mathcal{L}_G = \{E\}$ be the language of graphs with one binary relation symbol E .

Simple graphs are formalized by the axioms

- $\forall x \neg E(x, x)$
- $\forall x, y (E(x, y) \rightarrow E(y, x))$

We denote the theory of simple graphs by T_G

Exercise 0.1. Verify that

$$\overline{K} = \{\mathcal{G} : \mathcal{G} \models T_G, \mathcal{G} \text{ finite}\}$$

is an amalgamation class with countably many isomorphism types.

It follows that $\overline{K}$ has a Fraïssé limit. What does this limit look like?

Random graphs

We add the following **extension axioms**:

$$\sigma_{n,m} \equiv \forall x_1, \dots, x_n \forall y_1, \dots, y_m \left(\bigwedge_{i=1}^n \bigwedge_{j=1}^m x_i \neq y_j \rightarrow \exists z \left(\bigwedge_{i=1}^n (z \neq x_i \wedge E(z, x_i)) \wedge \bigwedge_{j=1}^m (z \neq y_j \wedge \neg E(z, y_j)) \right) \right)$$

Let

$$T_{RG} = T_G \cup \{\exists x, y (x \neq y)\} \cup \{\sigma_{n,m} : n, m \geq 1\}$$

Exercise 0.2. Show that every countable model of T_{RG} is homogeneous.

Exercise 0.3. Show that if $\mathcal{R}$ is a model of T_{RG} , $\text{age}(\mathcal{R}) = \overline{K}$.

It follows from Fraïssé's Theorem that T_{RG} has a countable model $\mathcal{R}$, and that this model is unique up to isomorphism.

Exercise 0.4. Show that T_{RG} is complete.

$\mathcal{R}$ is called the **random graph** and T_{RG} the *theory of the random graph*. Why? Let G_N be the set of all simple graphs on the vertex set $\{1, \dots, N\}$. Put a probability distribution on G_N by giving every graph the same probability. (Alternatively, we could independently assign edges to any vertex pair with probability $1/2$. This is called the **Erdős-Rényi model**.) Given an $\mathcal{L}_G$ -sentence σ , let $\mathbb{P}_N(\sigma)$ be the probability that σ holds for a random graph in G_N , i.e.

$$\mathbb{P}_N(\sigma) = \frac{|\{\mathcal{G} \in G_N : \mathcal{G} \models \sigma\}|}{|G_N|}.$$

It turns out, for large N , graphs will satisfy the extension axioms with high probability.

Exercise 0.5. Show that for any $n, m \geq 1$,

$$\lim_{N \rightarrow \infty} \mathbb{P}_N(\sigma_{n,m}) = 1.$$

This justifies referring to $\mathcal{R}$ as *the random graph*, since it can be seen as the “limit” of finite random graphs.

Further explorations

0-1 law for graphs

Theorem 0.1. For any $\mathcal{L}_G$ -sentence σ ,

$$\text{either } \lim_{N \rightarrow \infty} \mathbb{P}_N(\sigma) = 0 \text{ or } \lim_{N \rightarrow \infty} \mathbb{P}_N(\sigma) = 1.$$

Moreover, T_{RG} axiomatizes the theory

$$\{\sigma : \lim_{N \rightarrow \infty} \mathbb{P}_N(\sigma) = 1\}$$

and this theory is complete.

To prove the theorem, use that T_{RG} is complete together with Exercise 0.5. Give it a try!

Other Fraïssé classes

There are other classes of structures that are amalgamation classes, for example:

- finite fields of characteristic p
- finite groups
- (non-trivial) finite Boolean algebras

What kind of object is the Fraïssé limit in each case?